

in AI in the A&D domain also include research initiatives in the field of navigation techniques for drone swarms, cognitive electronic warfare, detection and signal processing in sensors, and dynamic spectrum allocation for radar/radio equipment in the congested radio frequency (RF) spectrums.

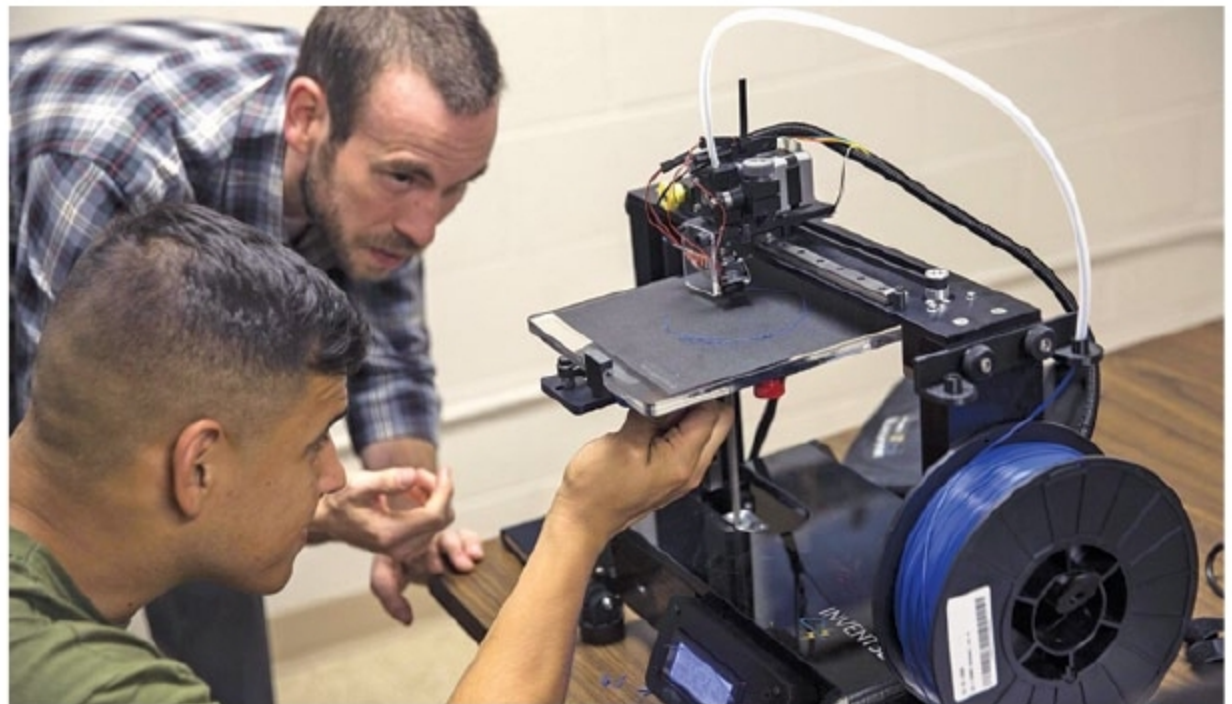
3D printing will gain importance as a low-cost parts manufacturing option, particularly in remote environments such as forward-deployed locations. As engineering improves either through component design or the way powders and filaments are developed, it will be utilised as a point to reduce cost over traditional fabrication methods.

Defence industry leaders are also investing in the development of new-generation diesel electric submarines with high endurance capability. The fuel cell technology will enable these submarines to generate power without surfacing to charge their batteries. Various companies, such as ThyssenKrupp Marine Systems (TKMS), Naval group and DRDO, are developing new-generation air-independent propulsion (AIP) technologies for that purpose.

Unmanned underwater vehicles (UUV) are on the way to show their genuinely disruptive potential as reconnaissance and weapon systems. With reports of Russia testing Poseidon system (earlier named Status-6) in December 2018, a nuclear powered-nuclear armed UUV with theoretically unlimited range and endurance, threat perception has become real. This will have an impact on the present development of such weapons, as well as defences against them, likely starting a new range of weapons.

The industry will see an accelerating development of counter-unmanned air systems (C-UAS). Vehicle-mounted C-UASes are still under testing and have limited usage; however, usage is expected to increase in the coming days.

Command, control, communications, computers, intelligence, surveillance and reconnaissance (C4ISR) is another domain that will undergo significant technological innovation



Use of 3D printing in defence (Credit: www.marines.mil)

and development. Influence of C4ISR will be seen in the following areas:

- Multifunction RF systems consolidating several RF functions like radar, electronic support, electronic attack and datalink communications within a shared set of electronics and antenna apertures using active electronically scanned array (AESA) technology
- Satellite constellation signifying a paradigm shift in space-based electro-optical and infrared (EO/IR) sensors
- Anti-access area denial (A2/AD) threats driving the development of aerial network alternatives to satellite communications (SATCOM)
- Use of gallium-nitride indicating a shift towards non-silica-based semiconductors, in the field of power electronics in defence applications, such as amplifiers for radar-based applications
- Application of short-wave infrared (SWIR) sensors for crucial surveillance and reconnaissance missions, as well as applications in medium-sized turrets, targeting pods and portable systems
- Cyber-security concerns impacting procurement schedules and boost funding, particularly of commercial off-the-shelf (COTS) wireless technology
- Tactical cloud-based applications reducing reliance on data centres, and offering a rapidly available and powerful battlefield cloud where tactical devices can be seamlessly connected

Focusing on Design in India

Despite huge opportunities, the Indian strategic electronics sector is still far from achieving self-reliance due to inadequate amalgamation of research and development (R&D) with manufacturing and absence of an integrated approach.

In the last few years, the defence sector has got a policy boost through the amendment of the defence procurement policy (DPP). DPP 2016 includes a special category called **Buy Indian - Indigenously Designed, Developed and Manufactured, or Buy Indian-IDDMM**, which will get the highest priority in the defence procurement process—above all other categories, namely, Buy and Make Indian, Buy and Make, and Buy Global, in that order.

Inclusion of this new procurement category—Buy Indian-IDDMM—will provide a greater thrust to Make in India initiative in defence production focusing on Design in India products and solutions.

This would promote in-house design capacity and higher localisation, strengthening the role of the domestic industry, especially the private sector, in defence production. However, it is to be noted that responsibility to prove an indigenous design rests with the industry, while the final approval would come from the government. This, in turn, would offer unique opportunities to companies that are operating or would like